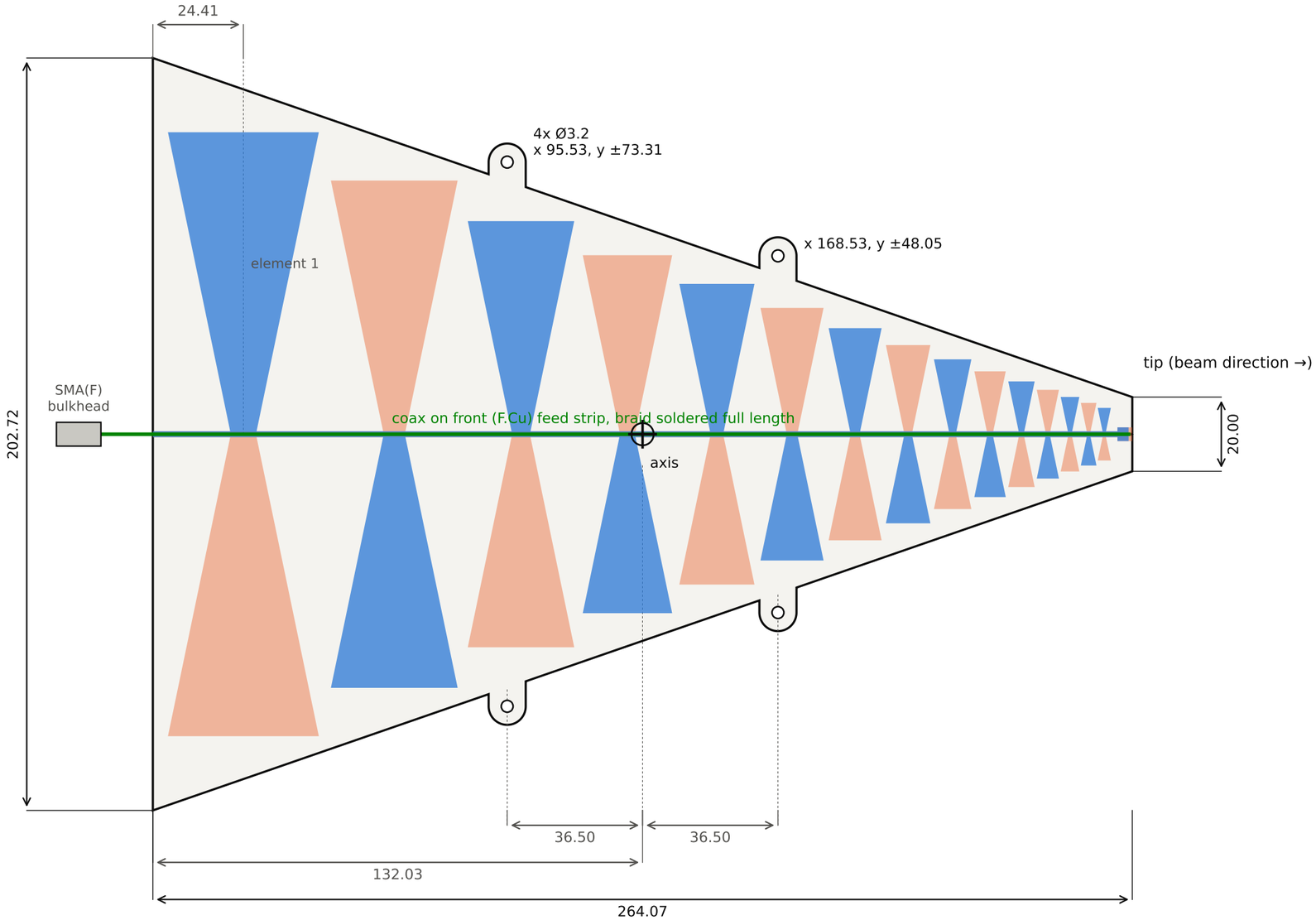
Log-periodic bow-tie array on RO4350B 20 mil — mechanical sheet (candidate, 2026-11-12, from the HFSS model; not a released drawing)

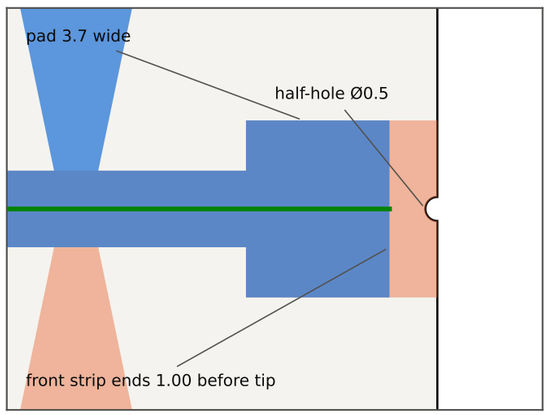
Front view (coax side). Blue = front copper, orange = back copper. Dimensions in mm.



Bow-tie elements (mm). Arm length is measured from the centreline; the back layer is the mirror image.

#	centre x	arm	end w	root w	front side
1	24.41	81.36	40.68	6.78	+y
2	65.09	68.34	34.17	5.70	-y
3	99.26	57.41	28.70	4.78	+y
4	127.96	48.22	24.11	4.02	-y
5	152.07	40.51	20.25	3.38	+y
6	172.33	34.03	17.01	2.84	-y
7	189.34	28.58	14.29	2.38	+y
8	203.63	24.01	12.00	2.00	-y
9	215.64	20.17	10.08	1.68	+y
10	225.72	16.94	8.47	1.41	-y
11	234.19	14.23	7.11	1.19	+y
12	241.30	11.95	5.98	1.00	-y
13	247.28	10.04	5.02	0.84	+y
14	252.30	8.43	4.22	0.70	-y
15	256.52	7.08	3.54	0.59	+y

Detail: tip (about 12:1)



MATERIAL Rogers RO4350B, 20 mil (0.508 mm) core, 1 oz (35 µm) copper both sides, no solder mask. Total thickness about 0.58 mm.

FEED Strips 1.60 mm wide on the centreline of both faces. Front strip 263.07 mm long; back strip runs to the tip. Matching pad at the tip: 3.70 mm wide.

COAX Taoglas CAB.01061: SMA(F) bulkhead + 300 mm of 1.37 mm micro-coax. Cut off the U.FL end, strip the jacket over the board length (braid 1.15 mm), solder the braid to the front strip. The SMA hangs off the base edge on the cable; it is not board-mounted.

TIP The 0.31 mm centre conductor leaves the coax 1.00 mm before the tip, wraps the tip edge through the Ø0.5 half-hole and is soldered to the back strip. Keep it as short as possible.

MOUNTING Four Ø3.2 holes (M3 clearance) on copper-free tabs 10 mm wide: x = 95.53, y = ±73.31 and x = 168.53, y = ±48.05 from the centreline. Turntable axis at x = 132.03 on the centreline. Use nylon screws. The tabs, holes, fixture, stiffener and SMA body are NOT in the simulated design.

SUPPORT GUIDANCE (not simulated) Keep all metal and solid plastic away from the feed strips and the element arms. Preferred: low-permittivity foam bonded to the back face, or a frame touching only the outline. Keep metal (motor, bearing, plate) well below the board; re-simulate with the fixture.

HEIGHT The soldered coax stands about 1.2 mm proud of the front face along the whole centreline (plus solder); the wire wrap adds about 0.3 mm beyond the tip edge and on the back face at the tip.

RF (simulated) First -10 dB crossing 0.685 GHz; S11 at or below -13 dB across the 902-928 MHz, 1.71-1.99 GHz, 2.4 GHz and 5.15-5.9 GHz bands; beam along +x (toward the tip).